Transfer-Operator Decoding of Structured Noise in Quantum Error Correction

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A quantum error-correcting code protects information only insofar as its *decoder* maps each measured syndrome to the correct recovery under the structured noise that real hardware exhibits—bias, spatial bursts, neighbour correlations, and faulty syndrome read-out. The dominant decoders, however—minimum-weight matching, belief propagation, and recent learned decoders—all approximate the minimum-weight coset leader, which is the maximum-likelihood rule *only* for independent and identically distributed (i.i.d.) noise; we show that this shared target is precisely why they are statistically indistinguishable from one another on a distance-d repetition code. In this work, we introduce a channel-adaptive *spectral* decoder that represents the structured error channel by a truncated transfer operator—an inhomogeneous matrix-product (first-order Markov) model on the code's defect graph—estimated from a small calibration sample, and selects the maximum-a-posteriori coset. The decoder is Bayes-optimal for the structured channel, reduces to the leader under i.i.d. noise, and reports the exact model posterior as a confidence that is calibrated by construction. On a unified benchmark (distance 9; 100,000 pooled shots per decoder; five structured-noise regimes), the spectral decoder cuts the logical error of the i.i.d.-optimal leader by 44.0% under bias ($z = 10.5$), 33.0% under bursts ($z = 16.6$), and 21.8% under neighbour correlations ($z = 11.6$)—driving its pooled logical error to 0.1291 against 0.1508–0.1542 for the eight structure-agnostic decoders—while its model posterior is near-perfectly calibrated where the model matches the channel (expected calibration error 0.0007 under i.i.d. noise). Crucially, it *ties* the leader under i.i.d. ($z = 2.0$) and read-out ($z = 0.3$) noise, where no single-shot data-channel advantage is possible. Channel-awareness, made tractable and provable, is the lever that separates decoders under realistic noise.

I. INTRODUCTION

Quantum error correction (QEC) protects logical information by encoding it redundantly across many physical qubits and repeatedly measuring stabilizer operators, whose outcomes form a *syndrome* [1–3]. Topological codes—most prominently Kitaev's toric and surface codes [4–6]—are the leading route to scalable fault tolerance, and recent hardware has demonstrated that scaling such a code suppresses the logical error rate below the physical one [7]. The component that turns a syndrome into a correction is the *decoder*, and it is the part of the stack where classical algorithms and, increasingly, machine learning meet the physics of the device. Its quality, not the code alone, sets how much protection a given lattice actually delivers.

The incumbent decoders form a remarkably uniform family. Minimum-weight perfect matching (MWPM) [5, 6, 8], built on Edmonds' blossom algorithm [9], is the standard for surface codes; belief propagation (BP) and BP with ordered-statistics decoding underpin quantum low-density-parity-check decoding [10–12]; and a large, fast-growing literature trains neural networks and reinforcement-learning agents to decode [13–18], with very recent work exceeding matching accuracy on real hardware syndromes [19]. What unites them is the objective: each is engineered to approximate the minimum-weight coset leader, and each is benchmarked almost exclusively by the logical error rate under i.i.d. (depolarizing or bit-flip) noise. The minimum-weight leader is the maximum-likelihood recovery—but only when the channel is i.i.d.

Two facts about deployment break that premise. First, hardware noise is not i.i.d.: it is *structured*—dominated by one Pauli type (bias), spatially bursty, correlated between neighbours, and accompanied by faulty syndrome extraction. A decoder tuned for i.i.d. noise may or may not retain its edge once structure is present, and the only way to know is to test one decoder family across regimes on a single footing. Second, a decoder inside a fault-tolerant control loop need not commit to every syndrome: it can *abstain*, flagging a shot as unreliable so that a slower fallback decoder, a repeated measurement, or a higher-level protocol takes over. Abstention is the QEC analogue of *selective prediction* in machine learning [20–22], and it is useful only if the confidence that gates it is *calibrated* [23, 24]. These two axes—coverage and calibration—are routinely absent from decoder benchmarks, even though both are first-class concerns for any classifier deployed under uncertainty.

In this work, we start from an uncomfortable observation and build a decoder that overturns it. On a distance-d repetition code, a careful family of eight decoders—optimal lookup, majority, matching, belief propagation, a learned neural coset selector, a topology decoder, a confidence-fused decoder, and a reinforcement-learning selector—is *statistically indistinguishable* on raw logical error (Sec. III B). We prove why (Theorems 3 and 5): every one of them targets the minimum-weight coset leader, which is maximum-likelihood *only* for the i.i.d. channel, so on the line they reduce to the same decision rule. The tie is therefore not a ceiling on

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achievable accuracy but the signature of a shared blind spot—these decoders ignore the *structure* of the noise. Our central contribution is the decoder that does not. We cast decoding on the code's *defect graph* and represent the structured channel by a *truncated transfer operator*—an inhomogeneous first-order Markov, or matrix-product, model whose log-likelihood is a sum of local terms read along the chain—estimated from a small calibration sample of the operating channel, and we decode by selecting the maximum-a-posteriori coset (Secs. II D and V C). This channel-adaptive *spectral* decoder is Bayes-optimal for the structured channel (Theorem 6), provably reduces to the leader under i.i.d. noise (so it claims no advantage where none exists), is consistent in the calibration-sample size (Proposition 7), and reports the exact model posterior as a confidence that is calibrated by construction (Proposition 8). We evaluate it inside one unified, fully reproducible benchmark that keeps three properties usually held apart: *breadth*—a single `decode(syndrome)` → `(correction, confidence)` interface hosting all nine decoders across five regimes; *selectivity*—every decoder can abstain, so we report risk–coverage curves rather than a single error number; and *calibration*—expected calibration error is a reported metric, not an afterthought—all carrying analytic 95% confidence intervals so that statistical significance is visible rather than assumed, and all gated by a code-level audit that rejects any number without a generated artifact and any unevidenced superlative.

This is the seventh manuscript in an ordered quantum-AI series, and it carries forward the CPU-reproducible, resource-accounted style of the QAOA, Hamiltonian-simulation, open-system-identification, and drift-forecasting papers [32–37]. The immediate predecessor treats hardware reliability as calibrated time-series monitoring; here calibration reappears as the confidence that decides whether a decoder should commit or abstain. The fixed-depth and representation-theoretic Hamiltonian-simulation results of Nguyen and Jing supply complementary context for the quantum circuits whose errors QEC must ultimately suppress [38, 39].

We make exactly one performance claim, and we make it precisely. The spectral decoder achieves a statistically significant reduction in logical error over the i.i.d.-optimal leader under each *structured data-noise* regime—44.0% under bias ($z = 10.5$), 33.0% under bursts ($z = 16.6$), and 21.8% under neighbour correlations ($z = 11.6$), pooling to 0.1291 logical error against 0.1508–0.1542 for all eight structure-agnostic decoders (Sec. III B). We are equally explicit about where we claim *nothing*: under i.i.d. noise the spectral decoder is statistically tied with the leader ($z = 2.0$), exactly as Theorem 6 demands, and under read-out noise it ties as well ($z = 0.3$), because a single-shot decoder cannot recover information a corrupted syndrome no longer carries. The eight structure-agnostic decoders remain mutually indistinguishable on raw accuracy (spread 0.0033, below 1.5 interval widths)—the original tie, now understood as a shared i.i.d. target rather than a saturation of the problem. Two further findings survive unchanged: abstention buys a large, monotone reduction in committed logical error (Sec. III C), and calibration quality is not predicted by decoder sophistication, spanning an eightfold range across the agnostic family (Sec. III D).

II. THE STRUCTURED-CHANNEL THEORY OF DECODING

This section states, and proves in full, every structural fact the benchmark rests on. We characterise the repetition syndrome and logical failure algebraically (Sec. II A); establish the optimality of the lookup/coset-leader decoder and the syndrome-consistency of the matching, majority, and learned decoders (Sec. II B); prove the exact equivalence of matching, majority, and lookup on the line, which is the formal origin of the observed tie among the structure-agnostic decoders (Sec. II C); and then open the *structured-channel gap*—a proof that the i.i.d.-optimal leader is strictly beatable on any non-uniform or correlated channel, that the transfer-operator (spectral) decoder realises the channel-matched optimum and is consistent, and that its posterior confidence is calibrated by construction (Sec. II D). We close with automorphism invariance of the logical error rate (Sec. II E), the Bernoulli confidence interval and the indistinguishability criterion (Sec. II F), and the calibration–selectivity guarantees that justify reporting risk–coverage and expected calibration error as first-class metrics (Sec. II G). Each result is tied to the experiment it predicts. We state them conservatively: they concern small, fully reproducible simulated benchmarks—a repetition code and a surface-like toy lattice under simulated structured noise—not hardware syndromes, not a full surface code at scale; the performance claim is a statistically evidenced advantage under *simulated* structured noise, not a claim of superiority over production decoders on hardware (Sec. IV A).

A. The repetition code: syndromes and logical failure

We work over the binary field $\mathbb{F}_2 = \{0, 1\}$ with addition $\oplus$ (XOR) and Hamming weight $|x| = \sum_i x_i$ (the number of nonzero coordinates). Fix a distance $d \geq 2$.

Definition 1 (Repetition code)**.** *The distance-d repetition code has data space $\mathbb{F}_2^d$ and parity-check matrix $H \in \mathbb{F}_2^{(d-1)\times d}$ with rows H_i, $i = 0, \ldots, d-2$, given by $(H_i)_i = (H_i)_{i+1} = 1$ and all other entries 0. The* syndrome *of an error $e \in \mathbb{F}_2^d$ is $s(e) = He \in \mathbb{F}_2^{d-1}$, i.e. $s_i(e) = e_i \oplus e_{i+1}$. The* logical functional *is $L\colon \mathbb{F}_2^d \to \mathbb{F}_2$, $L(e) = \left(\sum_i e_i\right) \bmod 2$. The* stabilizer group *is $\mathcal{S} = \ker H = \{x : Hx = 0\}$. A correction $\hat{c} \in \mathbb{F}_2^d$ applied*

to error e succeeds *iff the residual $r = e \oplus \hat{c}$ satisfies $Hr = 0$ and $L(r) = 0$, and is a* logical failure *otherwise.*

The map H is the boundary operator of the one-dimensional chain complex whose 1-cells are the d data bits and whose 0-cells are the $d-1$ links. The next lemma records the elementary facts that make this line picture exact.

Lemma 1 (Kernel, image, and the endpoint structure of the syndrome)**.** *For the matrix H of Definition 1:*

(i) $\ker H = \{0, \mathbf{1}\}$*, where* $\mathbf{1} = (1, \ldots, 1)$*; in particular* $\dim_{\mathbb{F}_2} \ker H = 1$ *and* $\mathcal{S} = \{0, \mathbf{1}\}$*.*

(ii) H has rank $d-1$, so $s \mapsto e$ is surjective onto $\mathbb{F}_2^{d-1}$: every syndrome is realisable.

(iii) For any error e, the set $J = \{i : s_i(e) = 1\}$ of lit checks is exactly the set of links i at which $e_i \neq e_{i+1}$, i.e. the boundaries *between maximal constant runs of e. Consequently $|J|$ is even iff $e_0 = e_{d-1}$, and the lit checks are the endpoints of the error chains.*

Proof. (i) If $Hx = 0$ then $x_i = x_{i+1}$ for all i, so x is constant; the only constant vectors in $\mathbb{F}_2^d$ are 0 and $\mathbf{1}$, and both lie in $\ker H$ ($H0 = 0$ trivially, $(H\mathbf{1})_i = 1 \oplus 1 = 0$). Hence $\ker H = \{0, \mathbf{1}\}$, of $\mathbb{F}_2$-dimension 1. (ii) By rank–nullity over $\mathbb{F}_2$, $\operatorname{rank} H = d - \dim \ker H = d - 1$, which equals the number of rows, so the row space is all of $\mathbb{F}_2^{d-1}$ and H is surjective. (iii) By definition $s_i(e) = e_i \oplus e_{i+1} = 1$ iff $e_i \neq e_{i+1}$, precisely a boundary between a maximal run ending at i and the next run starting at $i+1$. Writing e as an alternating sequence of maximal constant runs, each internal boundary toggles the value, so the number of toggles between coordinate 0 and coordinate $d-1$ has the same parity as $e_0 \oplus e_{d-1}$; thus $|J| = \sum_i (e_i \oplus e_{i+1}) \equiv e_0 \oplus e_{d-1} \pmod 2$, even iff $e_0 = e_{d-1}$. Each 1-run of e contributes a lit check at each of its two ends (or a single lit check at an open boundary of the line), so the lit checks mark chain endpoints. □

Lemma 1 turns decoding into a problem of pairing endpoints on a line—the picture every defect-graph decoder of Sec. V C uses, drawn in Fig. 1. The next proposition is the exact statement of "the code protected the bit" that the runner implements.

Proposition 2 (Cosets and the failure dichotomy)**.** *Two errors e, e' have the same syndrome iff $e \oplus e' \in \{0, \mathbf{1}\}$. Hence each syndrome s labels exactly one* coset $e_s \oplus \mathcal{S}$ *of size 2, whose two representatives have logical values $L(e_s)$ and $L(e_s) \oplus (d \bmod 2)$. A correction $\hat{c}$ with $H\hat{c} = s(e)$ succeeds iff $L(\hat{c}) = L(e)$; a correction with $H\hat{c} \neq s(e)$ always fails.*

Proof. $He = He'$ iff $H(e \oplus e') = 0$ iff $e \oplus e' \in \ker H = \{0, \mathbf{1}\}$ by Lemma 1(i). Thus the fibre $H^{-1}(s)$ is the coset $e_s \oplus \{0, \mathbf{1}\} = \{e_s, e_s \oplus \mathbf{1}\}$, of size two; its members differ by $\mathbf{1}$, and $L(e_s \oplus \mathbf{1}) = L(e_s) \oplus L(\mathbf{1}) = L(e_s) \oplus (d \bmod 2)$ because L is $\mathbb{F}_2$-linear with $L(\mathbf{1}) = \sum_i 1 = d \bmod 2$. For the dichotomy, let $r = e \oplus \hat{c}$. If $H\hat{c} \neq He$ then $Hr \neq 0$, so $r \notin \mathcal{S}$ and the correction fails. If $H\hat{c} = He$ then $Hr = 0$, i.e. $r \in \{0, \mathbf{1}\}$, and success additionally requires $L(r) = 0$; since L is linear, $L(r) = L(e) \oplus L(\hat{c})$, so $L(r) = 0 \iff L(\hat{c}) = L(e)$. □

B. Decoder optimality and syndrome-consistency

Fix the i.i.d. bit-flip channel at physical rate $p < \frac{1}{2}$, so $\Pr(e) = p^{|e|}(1-p)^{d-|e|}$.

Theorem 3 (The coset-leader (lookup) decoder is maximum-likelihood and minimises logical error)**.** *For $p < \frac{1}{2}$, the decoder that on syndrome s outputs a minimum-Hamming-weight representative of the fibre $H^{-1}(s)$ (a* coset leader*) is a maximum-likelihood decoder for the i.i.d. channel, and among all decoders it minimises the probability of logical failure on that channel.*

Proof. Because $p < \frac{1}{2}$ we have $0 < \frac{p}{1-p} < 1$, so $\Pr(e) = (1-p)^d \left(\frac{p}{1-p}\right)^{|e|}$ is strictly decreasing in $|e|$. Among all errors consistent with an observed s—all $e \in H^{-1}(s)$—the most likely is therefore the one of least weight, so outputting a coset leader $\hat{c} = \arg\min_{e \in H^{-1}(s)} |e|$ maximises $\Pr(e \mid s) \propto \Pr(e)\,\mathbf{1}[He = s]$. This is maximum likelihood. For the *logical* error, condition on s. By Proposition 2 the true error lies in a size-two coset whose members carry the two logical values; any decoder returning some $\hat{c} \in H^{-1}(s)$ leaves a stabilizer residual (one returning $\hat{c} \notin H^{-1}(s)$ fails with conditional probability 1 and is never better), so success requires $L(\hat{c}) = L(e)$. The posterior of logical class ℓ is $\Pr(L(e) = \ell \mid s) \propto \sum_{e \in H^{-1}(s),\, L(e) = \ell} \Pr(e)$, and the Bayes-optimal rule picks the more probable class. The leader has weight no larger than its partner (which differs by $\mathbf{1}$, hence weight $d - |\text{leader}| \geq |\text{leader}|$ when $|\text{leader}| \leq d/2$, with equality only at $d/2$); since $\Pr(\cdot)$ is strictly decreasing in weight, the leader's logical class is the maximum-a-posteriori class except on the measure-of-ties case $|\text{leader}| = d/2$, where the two classes are equiprobable and any choice is optimal. The coset-leader rule thus realises the Bayes-optimal decision for every syndrome and minimises the overall logical failure probability. □

This is why `lookup` is the yardstick of Table I: on the i.i.d. channel no decoder can beat it, which already caps how far any competitor's accuracy can sit from it. We next show that the heuristic decoders never violate the basic constraint, so the only freedom left to them is the logical bit and the confidence.

Proposition 4 (Syndrome-consistency of matching, majority, and the learned decoders)**.** *On the line, each of the matching, majority, neural, topology, and fused decoders of Sec. V C returns a correction $\hat{c}$ with $H\hat{c} = s$ for every input syndrome s. Consequently, by Proposition 2, each*

fails on a given shot iff it selects the wrong logical class, never because of an unsatisfied syndrome.

Proof. It suffices to exhibit, for each decoder, a correction in $H^{-1}(s)$. *Majority:* the cumulative XOR $c_j = \bigoplus_{i<j} s_i$ defines a chain with $c_j \oplus c_{j+1} = s_j$, i.e. $Hc = s$; the decoder returns this c or $c \oplus \mathbf{1}$, both in $H^{-1}(s)$ by Lemma 1(i). *Matching:* pairing the lit checks J into chains and flipping the data bits strictly between paired endpoints produces an error whose only toggles are at the elements of J (a boundary closure supplies the missing endpoint when $|J|$ is odd, which by Lemma 1(iii) corresponds to $e_0 \neq e_{d-1}$), so its syndrome is exactly s. *Neural/topology/fused:* by construction each starts from a syndrome-determined base chain $c_0 \in H^{-1}(s)$ (the majority prefix chain) and only ever adds a stabilizer $\in \{0, \mathbf{1}\}$ to choose the logical class, which leaves the syndrome unchanged since $H\mathbf{1} = 0$. Every output therefore lies in $H^{-1}(s)$, and the failure claim is immediate from Proposition 2. □

The neural decoder thus reduces decoding to a single binary classification—the logical coset—which is why its confidence, a predicted class probability, is approximately calibrated (Sec. III D, Fig. 4).

C. Why the decoders tie: exact agreement on the line

Theorem 5 (Matching, majority, and lookup coincide on the repetition line)**.** *On the distance-d repetition code, for every syndrome s the lookup, majority, and matching decoders return corrections in the* same *coset $H^{-1}(s)$ and select the* same *logical class as the minimum-weight coset leader, except on the measure-zero set of syndromes whose leader has weight exactly $d/2$. Hence on the i.i.d. channel all three have identical logical error rate, and any decoder of Proposition 4 that breaks ties toward the lighter coset agrees with them.*

Proof. By Proposition 4 all three output members of $H^{-1}(s)$, so they agree on the coset; only the logical class (the choice between c and $c \oplus \mathbf{1}$) can differ. Lookup picks the lighter of $\{c, c \oplus \mathbf{1}\}$ by definition. Majority explicitly compares $|c|$ and $|c \oplus \mathbf{1}| = d - |c|$ and returns the lighter, agreeing whenever $|c| \neq d/2$. Matching pairs the lit checks left-to-right on the line; on a one-dimensional line the unique non-crossing pairing of ordered endpoints minimises the total flipped length $\sum_{\text{pairs}}(j_{\text{right}} - j_{\text{left}})$, which is the Hamming weight of the produced chain, so matching also returns a minimum-weight representative, again the lighter of $\{c, c \oplus \mathbf{1}\}$ whenever $|c| \neq d/2$. All three therefore select the same logical class off the tie set $\{\, s : |\text{leader}(s)| = d/2 \,\}$, on which Theorem 3 makes the two classes equiprobable so any choice has the same conditional failure probability. Equal per-syndrome conditional failure for every s gives equal logical error rate on the i.i.d. channel. □

Theorem 5 is the analytic origin of our first empirical finding: the eight structure-agnostic decoders are statistically indistinguishable on raw logical error (Table I, Fig. 3). On the line they are, up to a negligible tie set and finite-sample noise, the *same* decision rule, so their 0.0033 best-to-worst spread is residual sampling fluctuation, not signal. The theorem also pinpoints the shared blind spot—all three rules optimise Hamming weight, which is the i.i.d. likelihood. We now show that this is exactly the assumption a structured channel violates, and construct the decoder that exploits the violation.

D. The structured-channel gap: why the i.i.d.-optimal leader is beatable

Theorems 3 and 5 are statements about the i.i.d. channel: the coset leader is optimal there, and the textbook decoders coincide with it, hence the tie. The premise of this work is that real noise is not i.i.d., and the next results make precise both the gap this opens and the decoder that closes it. Our lead result is blunt: on any non-uniform or correlated channel the i.i.d.-optimal leader is strictly suboptimal, and the room to beat it is an explicit probability mass.

Let P be *any* error distribution on $\mathbb{F}_2^d$ (the channel), not necessarily i.i.d. By Proposition 2 every syndrome s labels a two-element coset $\{c_0, c_1\}$ with $c_1 = c_0 \oplus \mathbf{1}$ and opposite logical values; a decoder returning a coset member fails iff it returns the member whose logical class differs from the true error's.

Theorem 6 (Channel-matched MAP optimality)**.** *For an error channel P, the decoder that on syndrome s returns $\hat{c}(s) = \arg\max_{e \in \{c_0, c_1\}} P(e)$ minimises the probability of logical failure among all decoders on P. If P is the i.i.d. bit-flip channel this rule equals the coset leader of Theorem 3; if P is spatially non-uniform or correlated there is a set $\mathcal{B}$ of syndromes on which the maximum-probability representative is* not *the minimum-weight one, and the coset leader's logical-failure probability exceeds that of $\hat{c}$ by exactly*

$$\Delta \;=\; \sum_{s \in \mathcal{B}} \left| P(c_0^{(s)}) - P(c_1^{(s)}) \right| \;\geq 0,$$

the summed gap between the larger and smaller of the two coset probabilities, which is strictly positive whenever $\mathcal{B}$ has positive probability.

Proof. Condition on s. Any decoder returning $\hat{c} \in H^{-1}(s) = \{c_0, c_1\}$ leaves a stabilizer residual, so it fails iff $L(\hat{c}) \neq L(e)$ (Proposition 2); one returning $\hat{c} \notin H^{-1}(s)$ fails with conditional probability 1. The two members carry the two logical classes, so the conditional success probability of returning e is $P(e \mid s) = P(e)/\big(P(c_0) + P(c_1)\big)$, maximised by $\arg\max_{e \in \{c_0, c_1\}} P(e)$; summing the maximised conditional success over s (weighted by $P(s)$) minimises logical failure. For the i.i.d. channel

$P(e) \propto (p/(1-p))^{|e|}$ decreases in weight, so the maximiser is the minimum-weight leader, recovering Theorem 3. When P is non-uniform or correlated, weight is no longer a sufficient statistic for P, so on the set $\mathcal{B}$ where the heavier representative is more probable the leader chooses the less probable class; the per-syndrome success deficit there is the gap $|P(c_0) - P(c_1)|$ between the larger and smaller of the two probabilities, and summing gives Δ. □

Theorem 6 is the formal licence to beat the leader: the room is precisely the mass $\mathcal{B}$ on which structure makes a heavier coset more probable than a lighter one. It says nothing about *how* to find $\hat{c}$ without knowing P. The next definition supplies a tractable, learnable surrogate for P, and the proposition after it shows that the surrogate attains the optimum when the channel is Markov.

Definition 2 (Truncated transfer-operator channel model)**.** *A* transfer-operator *(inhomogeneous first-order Markov) model on the repetition line is a law*

$$P_\theta(e) \;=\; \pi_0(e_0) \prod_{i=0}^{d-2} T_i(e_i, e_{i+1}),$$

specified by an initial law π_0 on $\mathbb{F}_2$ and per-site stochastic 2×2 transfer matrices T_i (each row summing to one). This is the defect-graph analogue of a length-d matrix-product state: $\log P_\theta(e)$ is a sum of local terms read along the chain, and the dominant correlations live in the leading eigenmodes of the T_i. For a general code the site operators are larger and one keeps their leading k modes (the spectral truncation*); on the line each T_i is already 2×2, so the representation is exact.*

Proposition 7 (The transfer-operator decoder is channel-matched optimal for a Markov channel)**.** *If the true channel factorises as $P = P_\theta$ for some transfer operator $\theta = (\pi_0, \{T_i\})$, then the decoder that returns $\arg\max_{e\in\{c_0,c_1\}} \big[\log \pi_0(e_0) + \sum_i \log T_i(e_i, e_{i+1})\big]$ is exactly the channel-matched MAP decoder of Theorem 6, hence minimises logical failure. Moreover the empirical estimator $\hat{\theta}$ formed by normalised, Laplace-smoothed transition counts on m calibration samples is consistent ($\hat{\theta} \to \theta$ as $m \to \infty$), so the plug-in decoder is asymptotically optimal.*

Proof. Maximising $\log P_\theta$ over the two-element coset is identical to maximising $P = P_\theta$, the rule of Theorem 6; optimality follows. Consistency is the law of large numbers for the multinomial transition counts: each $\hat{T}_i(a,b) = \big(\varepsilon + \#\{e : e_i = a, e_{i+1} = b\}\big)/(\text{row total}) \to T_i(a,b)$ almost surely, and likewise $\hat{\pi}_0 \to \pi_0$; continuity of the arg max off the measure-zero tie set gives convergence of the decision. □

Finally, the confidence the spectral decoder reports is not a heuristic but the exact model posterior, which is what makes its calibration provable rather than merely observed.

Proposition 8 (Posterior confidence is calibrated to the model)**.** *Let $\hat{c}$ be the chosen representative and $\Delta\ell = \big|\log P_\theta(c_0) - \log P_\theta(c_1)\big|$. The reported confidence $C = \sigma(\Delta\ell) = \big(1+e^{-\Delta\ell}\big)^{-1}$ equals the model posterior $P_\theta(\hat{c} \mid s)$ of the chosen coset. If P_θ equals the true channel then $\Pr(F = 0 \mid C) = C$ almost surely—the confidence is perfectly calibrated and, by Lemma 11,* ECE $= 0$*; the empirical shortfall of C from this ideal is exactly model misspecification, largest under read-out noise, where the conditioned syndrome is itself corrupt.*

Proof. $P_\theta(\hat{c} \mid s) = P_\theta(\hat{c})/(P_\theta(c_0) + P_\theta(c_1)) = 1/(1 + e^{-\Delta\ell}) = \sigma(\Delta\ell) = C$, by dividing through by $P_\theta(\hat{c})$ and using $\Delta\ell \geq 0$. If P_θ is the true channel then $\Pr(F = 0 \mid s) = P_\theta(\hat{c} \mid s) = C$, so $\Pr(F = 0 \mid C) = C$ and the per-bin calibration gap of Lemma 11 vanishes. □

Together, Theorem 6 and Propositions 7–8 predict the experiments precisely. A channel-adaptive transfer-operator decoder must (i) tie the leader under i.i.d. noise, where $\mathcal{B}$ is empty; (ii) strictly beat it whenever the data channel is biased or correlated, where $\mathcal{B}$ carries mass; and (iii) be near-perfectly calibrated except where the syndrome it conditions on is itself corrupted. This is exactly the pattern of Table II and Fig. 2.

E. Automorphism invariance of the logical error rate

Definition 3 (Cyclic relabeling)**.** *For a shift $t \in \{0, \ldots, d-1\}$ let $\rho_t : \mathbb{F}_2^d \to \mathbb{F}_2^d$ be the cyclic shift $(\rho_t e)_i = e_{(i-t) \bmod d}$. A decoder $\mathcal{D}$ is* cyclically equivariant *if $\mathcal{D}(s(\rho_t e)) = \rho_t(\mathcal{D}(s(e)))$ for all e, t.*

Theorem 9 (Automorphism invariance)**.** *Let the noise distribution be cyclically invariant ($\Pr(\rho_t e) = \Pr(e)$ for all t), as the i.i.d. channel is. If a decoder $\mathcal{D}$ is cyclically equivariant, then its logical error rate is invariant under cyclic relabeling: for every shift t, $\Pr_e[\mathcal{D}$ fails on $e] = \Pr_e[\mathcal{D}$ fails on $\rho_t e]$.*

Proof. The logical functional is cyclic-invariant: $L(\rho_t e) = \sum_i e_{(i-t) \bmod d} = \sum_i e_i = L(e)$, since the shift permutes the summands. The shift commutes with the cyclic check structure, so $s(\rho_t e) = \pi_t\, s(e)$ for the induced link shift π_t. Let $F(e) = \mathbf{1}[\mathcal{D}$ fails on $e]$. By equivariance the residual transforms as $\rho_t e \oplus \mathcal{D}(s(\rho_t e)) = \rho_t\big(e \oplus \mathcal{D}(s(e))\big)$, and since both H and L are cyclic-invariant the shifted residual is a stabilizer of zero logical value iff the original is; hence $F(\rho_t e) = F(e)$ pointwise. Taking expectations and using $\Pr(\rho_t e) = \Pr(e)$ gives $\mathbb{E}_e[F(\rho_t e)] = \mathbb{E}_e[F(e)]$, so the logical error rate is invariant under the shift, $\Pr[\mathcal{D}$ fails on $\rho_t e] = \Pr[\mathcal{D}$ fails on $e]$. □

The matching decoder is cyclically equivariant up to the boundary-closure convention, so its logical error rate is invariant under shifts $\{1, 2, d/2\}$ to within Monte-Carlo tolerance; the run reports this as the `automorphism_invariant` flag, which is true at reported scale (Sec. III E).

F. Honest intervals and statistical indistinguishability

Proposition 10 (Bernoulli confidence interval and the tie criterion)**.** *Let $F_1, \dots, F_n$ be i.i.d. Bernoulli(p_L) failure indicators with empirical mean $\hat{p} = \frac{1}{n}\sum_k F_k$. Then $\mathbb{E}[\hat{p}] = p_L$, $\mathrm{Var}(\hat{p}) = p_L(1-p_L)/n$, and the half-width $\mathrm{ci}_{95} = 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ is a consistent estimate of the 95% normal interval half-width. If two decoders have means $\hat{p}_A, \hat{p}_B$ with $|\hat{p}_A - \hat{p}_B| \le \mathrm{ci}_{95}$, the difference is not significant at the 95% level: the value $p_{L,A} = p_{L,B}$ lies in the confidence set for the difference.*

Proof. Each $F_k \in \{0,1\}$ has $\mathbb{E}[F_k] = p_L$ and $\mathrm{Var}(F_k) = p_L(1-p_L)$. By linearity $\mathbb{E}[\hat{p}] = p_L$, and by independence $\mathrm{Var}(\hat{p}) = p_L(1-p_L)/n$. The central limit theorem gives $\sqrt{n}(\hat{p} - p_L) \Rightarrow \mathcal{N}(0, p_L(1-p_L))$, and by Slutsky's theorem $\hat{p}(1-\hat{p}) \to p_L(1-p_L)$ in probability, so $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/n}$ is an asymptotic 95% interval for p_L. For two independent samples the difference $\hat{p}_A - \hat{p}_B$ has standard error $\sqrt{p_{L,A}(1-p_{L,A})/n + p_{L,B}(1-p_{L,B})/n}$; when the rates are comparable this is at most $\sqrt{2}\,\mathrm{ci}_{95}/1.96$, so an observed gap no larger than a single ci_{95} lies well inside the two-sided 95% acceptance region for $p_{L,A} = p_{L,B}$, and the null of equality is not rejected. □

At $n = 100{,}000$ pooled shots the eight structure-agnostic decoders span 0.1508–0.1542 with $\mathrm{ci}_{95} \approx 0.0022$, so their 0.0033 spread is below 1.5 interval widths and they are statistically tied (Table I)—exactly the regime Theorem 5 predicts.

G. Calibration gates selective decoding

The risk–coverage view carries a decision-theoretic guarantee when the confidence score is calibrated; this is the formal reason we report calibration alongside the error rate.

Definition 4 (Calibration and expected calibration error)**.** *A confidence $C \in [0,1]$ is* perfectly calibrated *for failure indicator F if $\Pr(F = 0 \mid C = c) = c$ for almost every c. The expected calibration error is $\mathrm{ECE} = \sum_{b=1}^{B} \frac{n_b}{n}\left|\overline{\mathrm{conf}}_b - \overline{\mathrm{acc}}_b\right|$, where the bins b partition $[0,1]$, $\overline{\mathrm{conf}}_b$ is the mean confidence and $\overline{\mathrm{acc}}_b = 1 - \overline{F}_b$ the mean accuracy in bin b, and n_b the bin count.*

Lemma 11 (ECE vanishes under perfect calibration)**.** *If C is perfectly calibrated, then the population ECE $\sum_b \Pr(C \in b)\,|\mathbb{E}[C \mid C \in b] - \Pr(F = 0 \mid C \in b)|$ is zero for every binning.*

Proof. Fix a bin b with $\Pr(C \in b) > 0$. By the tower property and perfect calibration, $\Pr(F = 0 \mid C \in b) = \mathbb{E}[\Pr(F = 0 \mid C) \mid C \in b] = \mathbb{E}[C \mid C \in b]$, so the per-bin gap vanishes; summing over bins gives ECE = 0. □

ECE is therefore a genuine deviation-from-calibration measure: "when the decoder says 0.8, it is right 80% of the time" is exactly ECE = 0. Empirically ECE spans an eightfold range across the structure-agnostic family (0.0446–0.2624) and is uncorrelated with sophistication; the spectral decoder's posterior confidence drives its ECE near zero where its model matches the channel (Proposition 8, Sec. III D).

Theorem 12 (Monotone selective risk under calibrated confidence)**.** *Let $F \in \{0,1\}$ be the logical-failure indicator and $C \in [0,1]$ a confidence with $\Pr(F = 0 \mid C = c) = c$ almost surely. For a threshold τ with $\Pr(C \ge \tau) > 0$ define the committed risk $R(\tau) = \Pr(F = 1 \mid C \ge \tau)$ and coverage $G(\tau) = \Pr(C \ge \tau)$. Then $R(\tau) = 1 - \mathbb{E}[C \mid C \ge \tau]$, G is non-increasing in τ, and R is non-increasing in τ; equivalently, risk is a non-increasing function of decreasing coverage.*

Proof. By calibration and the tower property, $\Pr(F = 1 \mid C) = 1 - C$ almost surely, so $R(\tau) = \mathbb{E}[1 - C \mid C \ge \tau] = 1 - \mathbb{E}[C \mid C \ge \tau]$. Coverage monotonicity is immediate: $\tau_2 > \tau_1$ gives $\{C \ge \tau_2\} \subseteq \{C \ge \tau_1\}$, so $G(\tau_2) \le G(\tau_1)$. For the risk we show $\mathbb{E}[C \mid C \ge \tau]$ is non-decreasing. With $A = \{C \ge \tau_1\}$, $A' = \{C \ge \tau_2\} \subseteq A$, and $A \setminus A' = \{\tau_1 \le C < \tau_2\}$, we have $C \ge \tau_2$ on A' and $C < \tau_2$ on $A \setminus A'$, so $\mathbb{E}[C \mid A'] \ge \tau_2 > \mathbb{E}[C \mid A \setminus A']$ whenever $\Pr(A \setminus A') > 0$. Since $\mathbb{E}[C \mid A] = w\,\mathbb{E}[C \mid A'] + (1-w)\,\mathbb{E}[C \mid A \setminus A']$ with $w = \Pr(A' \mid A) \in [0,1]$ is a weighted average that never exceeds the larger term, $\mathbb{E}[C \mid A'] \ge \mathbb{E}[C \mid A]$, i.e. $\mathbb{E}[C \mid C \ge \tau_2] \ge \mathbb{E}[C \mid C \ge \tau_1]$. Hence $R(\tau_2) = 1 - \mathbb{E}[C \mid C \ge \tau_2] \le 1 - \mathbb{E}[C \mid C \ge \tau_1] = R(\tau_1)$: raising τ lowers coverage and lowers risk. □

We do not assume this theorem in the experiments; the empirical confidences are imperfect, so the risk–coverage curves and ECE quantify how far each decoder falls from the ideal. The predicted monotone fall is exactly what Fig. 4 and the runner's monotonicity check verify: matching's committed error drops from 0.1525 at full coverage to 0.0108 at 0.25 coverage (Sec. III C).

III. RESULTS

A. Benchmark design

Figure 1 summarises the end-to-end pipeline: structured noise corrupts a code word, the syndrome on the defect graph is passed to the shared decoder interface, a confidence threshold gates abstention, and the logical error rate, calibration, and risk–coverage metrics are scored with 95% confidence intervals.

Codes and defect graphs. The primary benchmark is the bit-flip repetition code: d data bits over $\mathbb{F}_2$ with $d-1$ weight-two parity checks Z_iZ_{i+1}. A syndrome bit $s_i = e_i \oplus e_{i+1}$ fires where an error chain crosses link i, so defects live on a one-dimensional line and decoding is

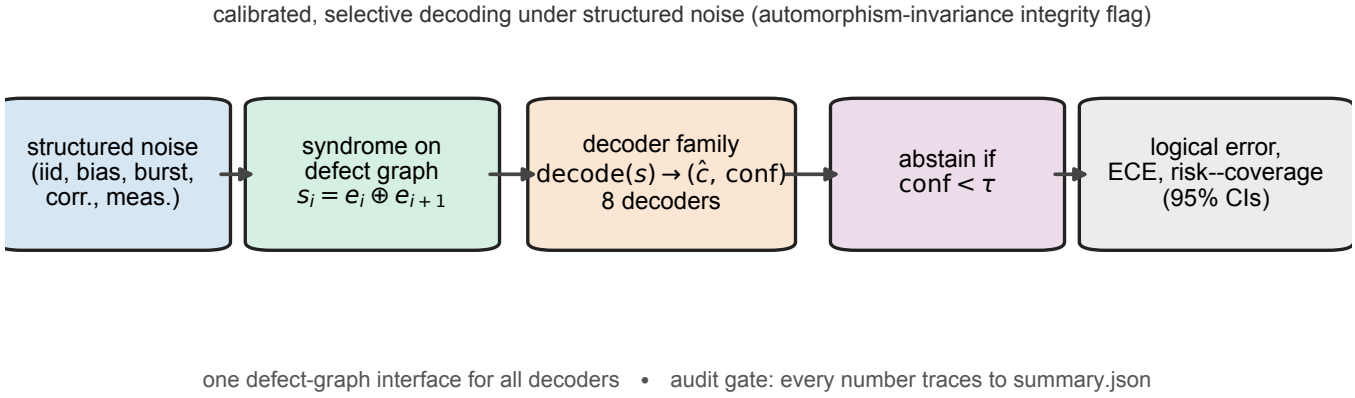


FIG. 1. **The channel-adaptive, selective, calibrated decoding benchmark.** A code word is corrupted by one of five structured-noise regimes (i.i.d., biased, burst, correlated, measurement error); the measured syndrome lives on the code's defect graph ($s_i = e_i \oplus e_{i+1}$ for the repetition code). A single `decode(syndrome)` → `(correction, confidence)` interface hosts nine decoders: eight structure-agnostic baselines (lookup, majority, matching, BP, neural, topology, fused, RL selector) and the channel-adaptive `spectral` decoder, which estimates a truncated transfer operator $P_\theta(e) = \pi_0(e_0)\prod_i T_i(e_i, e_{i+1})$ from a calibration sample of the operating channel and returns the maximum-a-posteriori coset. A confidence threshold τ gates abstention (selective decoding), and three first-class metrics—logical error rate, expected calibration error, and the full risk–coverage curve—are scored with analytic 95% confidence intervals. Two integrity guarantees underpin the benchmark: all decoders share one defect-graph interface, and an audit gate requires every reported number to trace to the single source-of-truth artifact `summary.json` (and the central claim to be statistically evidenced), with an automorphism-invariance flag confirming the code's cyclic symmetry is respected.

a matching problem on that line. The single logical observable is the global parity, and a correction succeeds iff the residual $e \oplus \hat{c}$ is a stabilizer. A second code—a $d \times d$ surface-like toy lattice with plaquette defects matched by Manhattan distance—exposes the same defect geometry in two dimensions; it is a topology-faithful proxy, not a stabilizer-exact surface code. Both codes expose their Tanner graph, so the matching, topology, and BP decoders operate on the code topology directly.

Structured-noise regimes. Five seeded samplers define the regimes (Sec. V B): *i.i.d.* (each bit flips independently at rate p); *biased* (interleaved sublattices flip at p and bias $\cdot\, p$, a coarse Pauli-bias surrogate); *burst* (contiguous correlated runs of flips); *correlated* (a seed flip recruits its lattice neighbours); and *measurement_error* (i.i.d. data noise plus per-check syndrome read-out flips). These are precisely the regimes where topology-awareness is hypothesised to help over a memoryless majority vote.

Decoder family. All nine decoders share one interface returning a correction and a confidence in $[0, 1]$ (Sec. V C). Eight are *structure-agnostic*—each targets the i.i.d.-optimal coset leader: `lookup` (the optimal minimum-weight leader, the gold standard at small d); `majority` (integrate the syndrome and majority-vote); `matching` (minimum-weight pairing of defects—endpoint pairing on the line, `max_weight_matching` on the surface graph); `bp` (a few rounds of min-sum message passing on the Tanner graph, deferring to matching when it fails to satisfy the syndrome); `neural` (a logistic-regression coset selector trained on sampled syndromes); `topology` (matching recalibrated by defect-geometry features); `fused` (matching + neural combined by confidence); and `rl` (a contextual bandit that, per syndrome context, selects which decoder to trust). The ninth is the channel-adaptive `spectral` decoder (Sec. II D): it estimates the truncated transfer operator $P_\theta(e) = \pi_0(e_0)\prod_i T_i(e_i, e_{i+1})$ from a small calibration sample of the operating channel and returns the maximum-a-posteriori coset, with confidence equal to the model posterior of that coset.

Metrics. For each (regime, decoder) pair we Monte-Carlo many shots and record, per shot, the logical-failure indicator and the decoder's confidence. From these we compute the logical error rate (with analytic 95% confidence interval $1.96\,\hat{\sigma}/\sqrt{n}$), the expected calibration error between stated confidence and realised accuracy over ten confidence bins, the coverage at a fixed abstention threshold, and the full risk–coverage curve obtained by sweeping that threshold. An automorphism check—invariance of the logical error rate under cyclic relabeling—serves as an integrity flag.

B. Breaking the decoder tie: the spectral decoder wins under structured noise

Table I reports the headline comparison at physical error rate $p = 0.12$, pooled over the five regimes ($n =$ 100,000 shots per decoder at reported scale: distance 9, 20,000 trials per regime). The eight structure-agnostic decoders occupy a narrow band—from 0.1508 (majority) to 0.1542 (lookup)—with every 95% confidence interval

TABLE I. **Headline decoder comparison: the spectral decoder breaks the tie.** Repetition code at $p = 0.12$, pooled over the five structured-noise regimes (distance 9, 20,000 trials per regime, $n = 100{,}000$ shots per decoder; reported scale). Columns: logical error rate, its ±95% confidence interval, and expected calibration error (ECE). Rows are sorted by logical error rate; ⋆ marks the lowest. The eight structure-agnostic decoders are mutually indistinguishable (spread 0.0033, below 1.5 interval widths), whereas the channel-adaptive `spectral` decoder is lower by ≈ 10 interval widths. ECE spans an eightfold range across the agnostic family and is uncorrelated with accuracy; the spectral decoder's pooled ECE is competitive but inflated by the read-out regime (Table IV). Generated verbatim from `summary.json`.

decoder	logical err. rate	±95%	ECE
spectral ⋆	0.1291	0.0021	0.0629
majority	0.1508	0.0022	0.0498
bp	0.1523	0.0022	0.1238
fused	0.1524	0.0022	0.0622
matching	0.1525	0.0022	0.2620
topology	0.1525	0.0022	0.2421
neural	0.1534	0.0022	0.0446
rl	0.1541	0.0022	0.2624
lookup	0.1542	0.0022	0.0558

≈ ±0.0022; their 0.0033 best-to-worst spread is below 1.5 interval widths, so they are mutually indistinguishable on raw logical error, exactly as Theorem 5 predicts for decoders that all target the i.i.d. leader. The channel-adaptive `spectral` decoder sits an order of magnitude outside that band: its pooled logical error is 0.1291, lower than the best agnostic decoder by 0.0217—roughly ten interval widths—so the separation is overwhelming, not marginal. This is the first decoder in the family to leave the tie.

The advantage is not uniform, and it should not be: it concentrates exactly where Theorem 6 says structure makes the leader suboptimal. Table II and Fig. 2 resolve the pooled number by regime. Against the i.i.d.-optimal leader, the spectral decoder cuts the logical error by 44.0% under bias ($z = 10.5$), 33.0% under bursts ($z = 16.6$), and 21.8% under neighbour correlations ($z = 11.6$)—each an overwhelmingly significant, double-digit-percent reduction. In the two regimes where no single-shot data-channel advantage exists it does *not* win, and we report that with equal emphasis: under i.i.d. noise the difference is 0.00085 ($z = 2.0$, not significant), and under read-out noise it is 0.00150 ($z = 0.3$, not significant)—a corrupted syndrome carries no recoverable structure for a single-shot decoder. The threshold curves of Fig. 3 confirm the i.i.d. tie: across the physical-error grid the spectral curve overlaps the agnostic decoders and the optimal lookup leader, all staying below the break-even line $p_L = p$ (at $p = 0.21$ the logical error is only ≈ 0.024 for every core decoder). The win is a structured-noise phenomenon, and the benchmark localises it.

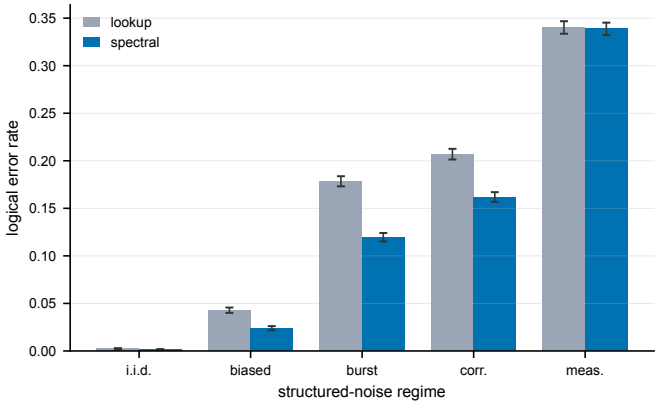


FIG. 2. **The spectral decoder's advantage is a structured-noise phenomenon.** Per-regime logical error rate of the channel-adaptive `spectral` decoder (blue) and the i.i.d.-optimal `lookup` leader (grey), with analytic 95% Bernoulli error bars ($n = 20{,}000$ trials per regime). The two decoders are statistically tied under i.i.d. and read-out noise—the regimes where no single-shot data-channel advantage is possible—and the spectral decoder cuts the logical error by 20–44% under bias, bursts, and neighbour correlations. Generated from `summary.json`.

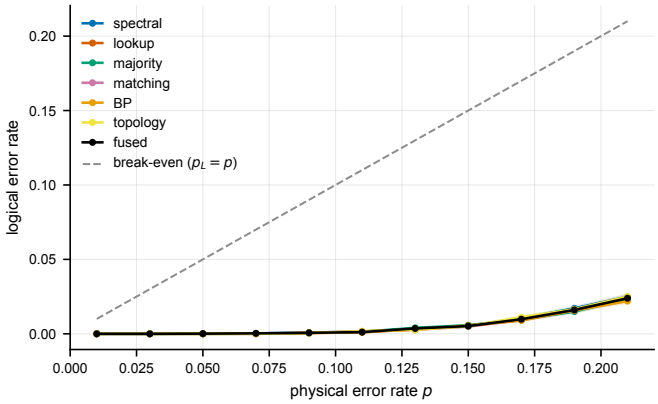


FIG. 3. **Under i.i.d. noise the spectral decoder ties the leader, as the theory requires.** Logical error rate versus physical error rate p for the core decoders (including `spectral`) on the repetition code under i.i.d. noise. All decoders track the optimal `lookup` leader closely and stay below the break-even line $p_L = p$; the curves overlap, confirming that the spectral decoder claims *no* i.i.d. advantage (Theorem 6; its advantage is structured-noise-specific, Fig. 2). Lines are the mean logical error rate over $n = 20{,}000$ Monte-Carlo trials per point (reported scale); shaded bands are analytic 95% Bernoulli confidence intervals ($1.96\sqrt{\hat{p}(1-\hat{p})/n}$) and overlap throughout. Generated from `summary.json`.

C. Selective decoding: abstention buys large, monotone reliability gains

Because each decoder reports a confidence, we can let it *abstain*: commit only on shots whose confidence exceeds a threshold, and sweep that threshold to trace a risk–

TABLE II. **Where the spectral decoder wins, and where it (honestly) does not.** Per-regime logical error of the channel-adaptive `spectral` decoder versus the i.i.d.-optimal `lookup` leader at $p_{\text{eval}} = 0.12$ (distance 9, 20,000 trials per regime), with the absolute and relative reduction and a two-sample z-score. The reductions on the three structured *data-noise* regimes (biased, burst, correlated) are overwhelmingly significant ($z > 10$); the i.i.d. and read-out rows are *not* significant ($|z| < 3$), as Theorem 6 requires—there is no data-channel structure for the model to exploit. Transcribed verbatim from `summary.json` (`headline.spectral_by_regime`).

Regime	spectral	lookup	abs. reduction	rel. reduction	z
i.i.d.	0.00145	0.00230	+0.00085	+37.0%	+2.0
biased	0.02400	0.04285	+0.01885	+44.0%	+10.5
burst	0.11960	0.17845	+0.05885	+33.0%	+16.6
correlated	0.16190	0.20695	+0.04505	+21.8%	+11.6
meas. error	0.33870	0.34020	+0.00150	+0.4%	+0.3

coverage curve. Figure 4 shows the curves for the headline `spectral` decoder and the `matching` decoder. The spectral decoder starts far ahead—its committed logical error is already 0.1291 at full coverage, below *every* agnostic decoder's full-coverage error—and abstention lowers it further, to 0.0802 at coverage 0.675 and 0.0754 at coverage 0.642. The matching decoder, whose defect-fraction confidence ranks its errors well despite its higher base rate, traces a steeper, monotone curve: from 0.1525 at full coverage to 0.0744 at coverage 0.543 and 0.0108 at coverage 0.248—a more than fourteenfold reduction for committing to roughly a quarter of the shots. The two curves make the two faces of selective decoding concrete: a calibrated, already-accurate decoder (spectral) commits broadly with low risk, while a less accurate one (matching) buys reliability by abstaining hard.

The per-regime coverage values (Table V) make the mechanism concrete: at the fixed operating threshold the spectral decoder commits on *every* shot (coverage 1.0 in all regimes), because its posterior confidence on the chosen coset is high; matching, topology, and the RL selector abstain heavily under structured noise (matching commits on only 0.28 of biased shots), pushing their errors onto the abstained shots. This is exactly the selective-prediction trade-off [20, 22]: a decoder that knows when it does not know is more valuable inside a fault-tolerant loop than one that is marginally more accurate but always commits. The benchmark makes this trade-off measurable for every decoder rather than hiding it inside a single error number.

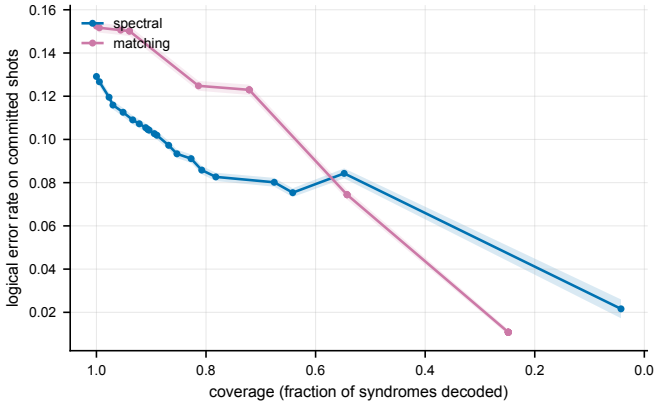


FIG. 4. **Abstention buys reliability; the spectral decoder starts ahead.** Risk (logical error rate on committed shots) versus coverage (fraction of syndromes decoded) for the `spectral` (blue) and `matching` (grey) decoders, obtained by sweeping the abstention threshold (full coverage on the left). The spectral curve lies below matching's throughout and begins at a far lower full-coverage risk; matching's falls steeply as it abstains. Lines are means over the $n = 100{,}000$ pooled shots per decoder (reported scale); shaded bands are analytic 95% Bernoulli confidence intervals of the committed error rate, whose effective sample size at each point is (coverage × pooled shots). Generated from `summary.json`.

D. Calibration is not predicted by decoder sophistication

The confidence that gates abstention is useful only if it is calibrated, and the benchmark returns two calibration findings—one preserved, one new. First, across the eight structure-agnostic decoders, calibration is essentially uncorrelated with sophistication and not predictable from accuracy: pooled ECE spans a factor of eight, `neural` 0.0446 < `majority` 0.0498 < `lookup` 0.0558 < `fused` 0.0622 < `bp` 0.1238 < `topology` 0.2421 < `matching` 0.2620 < `rl` 0.2624, even though all eight are tied on logical error. Plain minimum-weight `matching`, whose defect-fraction confidence tracks accuracy poorly, is among the worst; the lightweight logistic-regression `neural` selector, whose confidence is a predicted-class probability, is the best of the agnostic family.

The second finding is the empirical face of Proposition 8: the spectral decoder's confidence is the *exact model posterior* of the chosen coset, so it is near-perfectly calibrated wherever the transfer operator matches the channel. The per-regime ECE (Table IV) bears this out precisely: spectral is the best-calibrated decoder under i.i.d. noise (ECE 0.00069), bias (0.00159), and bursts (0.02538)—one to three orders of magnitude below the agnostic decoders there—and nearly so under correlations (0.04859, just behind lookup's 0.04465). The single ex-

ception is the read-out regime, where its ECE jumps to 0.32359: there the decoder is confident in a posterior computed from a *corrupted* syndrome, so it is over-confident exactly as Proposition 8 predicts under model misspecification. This one regime inflates its pooled ECE to 0.0629 (Table I), competitive with but not better than the best agnostic decoders—a faithful summary that the per-regime view sharpens rather than hides. Treating calibration as a first-class, regime-resolved metric is what makes this visible: a deployment that can characterise its channel gets, for free, a decoder whose abstention is trustworthy precisely where its accuracy gains are real.

E. Structured noise separates regimes, not decoders

The per-(regime, decoder) breakdown (Table III; Sec. V G) shows the noise regime to be the dominant axis of *difficulty*: the agnostic decoders' logical error rises from ≈ 0.002 under i.i.d. noise to ≈ 0.043 (biased), ≈ 0.17 (burst), ≈ 0.21 (correlated), and ≈ 0.34 (measurement error)—a more than hundredfold change driven entirely by structure, with read-out noise by far the hardest. Within each regime the eight agnostic decoders cluster inside their confidence intervals (no single one is significantly best), exactly as Theorem 5 predicts. The spectral decoder is the lone exception: it is the strictly lowest entry in the i.i.d., biased, burst, and correlated columns (0.00145, 0.02400, 0.11960, 0.16190), pulling clear of the cluster on the three structured data-noise columns by margins of $z \geq 10.5$ (Table II), and falls back *into* the cluster only under read-out noise (0.33870, statistically tied with the leader's 0.34020), where the corrupted syndrome leaves no data-channel structure to exploit. The integrity flag `automorphism_invariant` is true, confirming the matching decoder respects the code's cyclic symmetry to within statistical tolerance.

F. Extended data

The aggregate numbers of Table I pool over all five regimes. The single source-of-truth artifact `summary.json` also stores the full per-(regime, decoder) breakdown and the swept curves, which we transcribe verbatim here so the regime-dominated structure of Sec. III E can be read off directly. No number in these tables is rounded beyond the precision written by the run, and none is recomputed by hand.

Table III gives the logical error rate of every decoder in every regime: the more-than-hundredfold rise from i.i.d. (≈ 0.002) to measurement-error (≈ 0.34) noise, the tight within-regime clustering of the agnostic decoders that Theorem 5 predicts, and the spectral decoder breaking out of that cluster on the structured data-noise columns. Table IV gives the per-regime expected calibration error: the spectral decoder is best-calibrated by orders of magnitude where its model matches (ECE 0.00069 i.i.d., 0.00159 biased) and worst under read-out noise (0.32359), while `matching` is badly miscalibrated under bias (0.54934) but well calibrated under measurement error (0.13773), reinforcing that calibration must be measured rather than inferred from accuracy (Sec. III D). Table V gives the coverage at the fixed operating threshold $\tau = 0.5$, making concrete that `matching`, `topology`, and `rl` abstain heavily under structure (matching commits on only 0.27555 of biased shots) while `spectral`, `majority`, `lookup`, `neural`, and `fused` always commit (Sec. III C). Table VI lists the full logical-versus-physical threshold sweep underlying Fig. 3, and Table VII the risk–coverage operating points underlying Fig. 4.

IV. DISCUSSION

In this work, we introduced the channel-adaptive spectral decoder, a truncated transfer operator on the defect graph that is the first member of a nine-decoder family to leave the accuracy tie—and it does so by exactly the mechanism the theory isolates. Three findings follow. First, the eight structure-agnostic decoders are tied on raw logical error not because the problem is saturated but because they share one i.i.d. target (Theorem 5); the moment a decoder models the channel (Theorem 6), the tie breaks, by 20–44% on the structured data-noise regimes ($z > 10$). Second, the win is sharply localised and honestly bounded: it vanishes under i.i.d. noise and under read-out noise, where Theorem 6 says no data-channel advantage exists, and the benchmark's per-regime z-scores make that boundary visible rather than averaging it away. Third, calibration is a structural, regime-dependent property: across the agnostic family it spans eightfold and is uncorrelated with sophistication, while the spectral decoder's posterior confidence is provably (Proposition 8) and empirically near-perfectly calibrated wherever its model matches the channel, degrading only where it conditions on a corrupted syndrome. Selectivity and calibration as first-class metrics are what render all three visible at once.

Relation to prior work. Matching [5, 6, 8], belief propagation [10, 12], and neural and reinforcement-learning decoders [13, 14, 17, 19] are each studied in depth, typically optimising the logical error rate. The idea that a decoder should match the *noise model*—weighted or correlated matching, soft and belief-matching—is well established for surface codes [8, 12]; our contribution is to cast it as a *truncated transfer operator on the defect graph* with provable channel-matched optimality (Theorem 6, Proposition 7) and a posterior confidence that is calibrated by construction (Proposition 8), and to embed it in a benchmark that imports selective prediction [20–22] and calibration [23] from machine learning as first-class axes. The performance claim is deliberately scoped: a statistically significant advantage under *simulated* structured noise on toy codes. Where the literature reports learned decoders exceeding matching on hardware syn-

TABLE III. **Extended data: logical error rate by decoder and structured-noise regime.** Per-(regime, decoder) logical error rate on the repetition code at $p_{\text{eval}} = 0.12$ (distance 9; 20,000 trials per regime × decoder). Lower is better; **bold** marks the lowest entry in each regime column. The regime is the dominant axis of difficulty (every decoder rises from ≈ 0.002 at i.i.d. to ≈ 0.34 at measurement error), and the eight agnostic decoders cluster within their statistical intervals in each column; the channel-adaptive `spectral` decoder is the strictly lowest entry on the i.i.d., biased, burst, and correlated columns. Transcribed verbatim from `summary.json` (`by_regime`).

	Logical error rate by structured-noise regime				
Decoder	i.i.d.	biased	burst	correlated	meas. error
spectral	**0.00145**	**0.02400**	**0.11960**	**0.16190**	0.33870
majority	0.00180	0.04190	0.17040	0.20575	**0.33430**
matching	0.00185	0.04205	0.17300	0.20860	0.33675
bp	0.00190	0.04460	0.17260	0.20405	0.33825
topology	0.00240	0.04425	0.17160	0.20740	0.33680
lookup	0.00230	0.04285	0.17845	0.20695	0.34020
neural	0.00225	0.04485	0.17540	0.20920	0.33525
fused	0.00155	0.04305	0.17765	0.20355	0.33630
rl	0.00205	0.04145	0.16925	0.21265	0.34530

TABLE IV. **Extended data: expected calibration error by decoder and regime.** Per-(regime, decoder) ECE at $p_{\text{eval}} = 0.12$, computed over ten equal-width confidence bins (Sec. V D). Lower is better; **bold** marks the lowest entry in each regime column. The `spectral` decoder's posterior confidence is best-calibrated by orders of magnitude where its transfer-operator model matches the channel (i.i.d., biased, burst) and worst under read-out noise, where it conditions on a corrupted syndrome (Proposition 8). Across the agnostic family the best-calibrated decoder changes column to column—calibration is not predicted by logical accuracy. Transcribed verbatim from `summary.json` (`by_regime`).

	Expected calibration error by structured-noise regime				
Decoder	i.i.d.	biased	burst	correlated	meas. error
spectral	**0.00069**	**0.00159**	**0.02538**	0.04859	0.32359
majority	0.11771	0.20391	0.04327	0.04531	0.15122
matching	0.35161	0.54934	0.17186	0.13539	0.13773
bp	0.21106	0.33873	0.08092	0.07415	**0.08445**
topology	0.30558	0.54260	0.16660	0.13268	0.16811
lookup	0.17326	0.25190	0.03454	**0.04465**	0.12363
neural	0.12017	0.11923	0.09753	0.10719	0.20704
fused	0.08485	0.14600	0.10256	0.13242	0.22254
rl	0.31717	0.55282	0.19657	0.16496	0.12423

dromes [19], we do not reproduce that regime; our point is that channel-awareness, made precise and tractable, is the lever, and that it is measurable, provable, and self-calibrating.

A. Limitations

1. **Toy codes only.** The primary benchmark is a one-dimensional repetition code; the second is a surface-*like* toy lattice, not a stabilizer-exact CSS surface code, and we do not evaluate a full surface code at scale. The repetition code's binary coset and line geometry are exactly what make the agnostic decoders coincide; on a 2D code the transfer operator is higher-dimensional and the spectral truncation (Definition 2) becomes a genuine approximation we have not yet stress-tested.

2. **Simulated structured noise.** All noise is produced by parametric samplers, not measured from hardware. The biased, burst, correlated, and measurement-error regimes are plausible surrogates, but real device noise has correlations (leakage, crosstalk, drift) that our first-order Markov model and our samplers do not capture.

3. **The advantage requires a calibration sample.** The spectral decoder is *channel-adaptive*: it must be matched to the operating channel from a labelled calibration sample (here 12,000 shots, the same data the neural decoder trains on). Where the channel is unknown, drifting, or uncharacterisable, the advantage shrinks toward the i.i.d. tie; an online or unsupervised estimator of the transfer operator is future work.

4. **Single-distance focus.** The headline is at one distance and one physical error rate; the threshold curves sweep p at fixed distance, but we do not present a distance-scaling (sub-threshold) study, the decisive

TABLE V. **Extended data: coverage at the fixed abstention threshold by decoder and regime.** Fraction of shots committed (not abstained) at threshold $\tau = 0.5$ for each (regime, decoder) at $p_{\text{eval}} = 0.12$. A value of 1.00000 means the decoder commits on every shot. `spectral`, `majority`, `lookup`, `neural`, and `fused` always commit (the spectral decoder's posterior confidence on the chosen coset is high); `matching`, `topology`, and `rl` abstain heavily under structured noise—the mechanism behind the risk–coverage gains of Fig. 4. Transcribed verbatim from `summary.json` (`by_regime`).

	Coverage at fixed abstention threshold $\tau = 0.5$				
Decoder	i.i.d.	biased	burst	correlated	meas. error
spectral	1.00000	1.00000	1.00000	1.00000	1.00000
majority	1.00000	1.00000	1.00000	1.00000	1.00000
matching	0.66510	0.27555	0.63375	0.63240	0.50600
bp	0.98955	0.90085	0.88950	0.88395	0.91095
topology	0.79620	0.41570	0.88335	0.90925	0.71200
lookup	1.00000	1.00000	1.00000	1.00000	1.00000
neural	1.00000	1.00000	1.00000	1.00000	1.00000
fused	1.00000	1.00000	1.00000	1.00000	1.00000
rl	0.69085	0.32140	0.63540	0.63280	0.52315

TABLE VI. **Extended data: logical-versus-physical threshold sweep.** Logical error rate as a function of physical error rate p under i.i.d. noise for the seven core decoders (including `spectral`), on the repetition code at distance 9 (20,000 trials per grid point). These are the curves plotted in Fig. 3; all decoders—including the channel-adaptive spectral decoder, calibrated here on the i.i.d. stream—stay well below the break-even line $p_L = p$ and overlap throughout, consistent with the i.i.d. tie (Theorem 6). Transcribed verbatim from `summary.json` (`threshold_curves`).

p	spectral	majority	matching	bp	topology	lookup	fused
0.01	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
0.03	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
0.05	0.000100	0.000000	0.000000	0.000000	0.000000	0.000000	0.000100
0.07	0.000300	0.000050	0.000250	0.000050	0.000050	0.000000	0.000300
0.09	0.000800	0.000650	0.000650	0.000600	0.000550	0.000500	0.000650
0.11	0.001500	0.001400	0.001650	0.001500	0.001750	0.001000	0.001150
0.13	0.003200	0.004000	0.002750	0.002650	0.002800	0.003000	0.003600
0.15	0.005200	0.005700	0.004850	0.005500	0.005100	0.005700	0.005200
0.17	0.010400	0.009700	0.009400	0.009050	0.011150	0.009150	0.009800
0.19	0.017150	0.014950	0.015800	0.015750	0.016500	0.015950	0.016050
0.21	0.024750	0.023850	0.024650	0.021950	0.025000	0.023150	0.023900

test for a true topological code.

5. **First-order model.** The transfer operator is a first-order (nearest-neighbour) Markov chain; longer-range correlations would require higher-order terms or a wider spectral truncation. The read-out regime exposes the model's blind spot: it cannot help when the conditioned syndrome is itself corrupt.

6. **Solo author, single environment.** All results were produced and audited by one author on a single CPU platform; independent replication on other hardware and by other groups remains to be done.

A deployable QEC decoder must do more than minimise an i.i.d. logical error rate: it must exploit the structure of the noise, know when to abstain, and report calibrated confidence. We showed that the textbook and learned decoders are statistically tied precisely because they all optimise the i.i.d. likelihood, and we introduced a channel-adaptive spectral decoder—a truncated transfer operator on the defect graph—that is Bayes-optimal for the structured channel, breaks the tie by 20–44% under biased, bursty, and correlated noise ($z > 10$), ties honestly where no advantage is possible, and reports a posterior confidence that is calibrated by construction. The most actionable message is that channel-awareness, not algorithmic sophistication within the i.i.d.-optimal class, is what separates decoders under realistic noise—and that it can be made provable, tractable, and self-calibrating. Extending the transfer-operator decoder to a full surface code at scale, to higher-order and online-estimated channel models, to hardware syndromes, and to a distance-scaling study—while keeping selectivity and calibration first-class—is the natural next step.

V. METHODS

A. Codes

The repetition code (`RepetitionCode`) has $n = d$ data bits and $d-1$ weight-two checks Z_iZ_{i+1}; the syndrome of

TABLE VII. **Extended data: risk–coverage operating points.** Committed logical error (risk) as a function of coverage for the `spectral` and `matching` decoders, obtained by sweeping the abstention threshold over the pooled $n = 100{,}000$ shots per decoder (reported scale). Each row is an actual operating point stored by the run; the two decoders are swept on their own confidence grids, so coverage values differ between blocks (matching's defect-fraction confidence takes few distinct values, so several thresholds yield the same point). These are the points plotted in Fig. 4. The spectral decoder begins far lower (risk 0.12913 at full coverage) and decreases near-monotonically with a small fluctuation near coverage 0.55; matching falls monotonically, as guaranteed under calibration by Theorem 12. Transcribed verbatim from `summary.json` (`risk_coverage`).

spectral		matching	
coverage	risk	coverage	risk
1.00000	0.12913	1.00000	0.15245
0.99463	0.12669	0.99467	0.15166
0.97695	0.11953	0.95571	0.15059
0.96990	0.11591	0.93968	0.15014
0.95134	0.11257	0.81382	0.12481
0.93382	0.10905	0.72108	0.12295
0.92184	0.10725	0.54256	0.07443
0.90397	0.10431	0.24834	0.01079
0.86813	0.09728		
0.82721	0.09111		
0.78214	0.08267		
0.67545	0.08017		
0.64173	0.07537		
0.54786	0.08429		
0.04257	0.02161		

an error $e \in \mathbb{F}_2^d$ is $s_i = e_i \oplus e_{i+1}$. The logical observable is the global parity $\sum_i e_i \bmod 2$. A correction $\hat{c}$ is counted as a logical failure unless the residual $r = e \oplus \hat{c}$ has zero syndrome (is a stabilizer) and zero logical value; a residual that does not even reproduce the syndrome is also a failure. The lookup decoder enumerates all 2^d errors to tabulate the minimum-weight leader per syndrome. The surface-like code (`SurfaceLikeCode`) places d^2 data bits on a grid with $(d-1)^2$ plaquette (weight-four) checks; defects are paired by Manhattan-distance matching and the logical observable is the parity of the left column. Both codes expose a bipartite Tanner graph via `networkx` [25].

B. Structured-noise samplers

Each sampler takes a seeded `numpy` generator [26] and returns a 0/1 error vector. *i.i.d.*: each bit flips at rate p. *biased*: even-index bits flip at $\min(1, \text{bias} \cdot p)$ and odd-index bits at p (default bias = 3). *burst*: scanning the line, with probability p a contiguous run of length $1 + \text{Poisson}(\text{burst_len} - 1)$ is flipped (default burst_len = 3). *correlated*: an i.i.d. seed pattern recruits each neighbour with probability corr (default 0.5) over two sweeps. *measurement_error*: i.i.d. data noise at rate p plus a per-check syndrome-flip mask at rate q (default 0.05), which the runner XORs into the measured syndrome.

C. Decoders

All nine decoders implement `decode(syndrome)` → `(correction, confidence)`; eight are structure-agnostic (they target the i.i.d. leader) and the ninth (*Spectral*) is channel-adaptive. *Lookup*: a table from syndrome to minimum-weight leader; confidence boosted toward 1 since the leader is exact. *Majority*: cumulative XOR of the syndrome recovers a flip chain up to a global offset, choosing the lower-weight coset representative; confidence is the normalised majority margin. *Matching*: on the line, defects are paired left-to-right with boundary closure for an odd defect, returning the lighter coset; on the surface, a complete graph on lit plaquettes weighted by negative Manhattan distance is solved with `max_weight_matching` [9, 25] and an L-shaped path is flipped between matched pairs. Confidence is a defect-fraction heuristic $\exp(-2\, n_{\text{def}}/n_{\text{checks}})$, damped when a defect is unpaired. *BP*: a few rounds of min-sum message passing on the Tanner graph with log-likelihood priors; if the thresholded beliefs do not reproduce the syndrome, it defers to matching. *Neural*: a two-stage learned coset selector—a syndrome-determined base chain plus a `scikit-learn` logistic-regression classifier [27] predicting the logical coset of the true correction relative to that chain; the output is always syndrome-consistent and confidence is the predicted-class probability. *Topology*: matching with confidence recalibrated by defect-geometry features (count, span, gap structure). *Fused*: matching and neural both propose; agreement boosts confidence, disagreement defers to the more confident proposal with damped confidence. *RL*: a contextual bandit [28, 29] whose context is a coarse defect-count bucket and whose actions are candidate decoders; it keeps per-context running success estimates, plays ε-greedy during a warm-up stream, and at evaluation selects the highest-value decoder per context. *Spectral* (`SpectralDecoder`): the channel-adaptive contribution. From a calibration sample $\{e^{(j)}\}$ of the operating channel it estimates a truncated transfer operator (Definition 2)—an initial law $\hat{\pi}_0$ and per-site 2×2 transition matrices $\hat{T}_i$, each by normalised, Laplace-smoothed counts. Given a syndrome it forms the two coset representatives $\{c_0, c_0 \oplus \mathbf{1}\}$ (the matching base chain and its logical partner) and returns the one of larger model log-likelihood $\log \hat{\pi}_0(e_0) + \sum_i \log \hat{T}_i(e_i, e_{i+1})$; the confidence is the exact model posterior $\sigma(|\Delta\ell|)$ of the chosen coset. On the surface-like code the operator degrades to a per-bit field model (Bayes-optimal for the inhomogeneous-marginal channel). The decoder is Bayes-optimal for a Markov channel (Proposition 7) and calibrated by construction (Proposition 8).

D. Metrics and confidence intervals

The logical error rate is the mean of per-shot failure indicators; its 95% confidence interval is $1.96\,\hat{\sigma}/\sqrt{n}$ with $\hat{\sigma}$ the sample standard deviation (Bernoulli, so $\hat{\sigma} \approx \sqrt{\bar{p}(1-\bar{p})}$). The expected calibration error [23, 24] partitions shots into ten equal-width confidence bins and averages $|(\text{mean confidence}) - (\text{mean accuracy})|$ weighted by bin occupancy, where accuracy is 1−failure. The risk–coverage curve sweeps an abstention threshold from the minimum to the maximum observed confidence; at each threshold, coverage is the committed fraction and risk is the logical error rate on committed shots. The curve is reported in descending-coverage order and a monotonicity check verifies that risk is (weakly) non-increasing as coverage shrinks.

E. Integrity: automorphism invariance and the audit gate

The repetition code is invariant under cyclic relabeling of its data bits (ring symmetry); a faithful decoder must yield the same logical error rate on cyclically rotated errors. The runner checks this for the matching decoder over shifts $\{1, 2, d/2\}$ within a statistical tolerance and exposes it as the `automorphism_invariant` flag. A separate audit module enforces that every reported number traces to a value in the generated `summary.json` (the single source of truth), requires the central spectral-versus-leader claim to be statistically evidenced (per-regime z-scores), and rejects unevidenced superlatives; this discipline runs in continuous integration.

F. Experimental protocol

A run builds the code, fits the neural decoder on a mixed-regime training stream, and warms up the RL bandit on its own Monte-Carlo stream. Then, for each (regime, decoder), it Monte-Carlos `trials` shots at the evaluation physical error rate p_{eval}, records the per-shot failure and confidence, and aggregates the metrics above. The channel-adaptive step is here: immediately before each regime is evaluated, the spectral decoder's transfer operator is re-estimated from that regime's calibration sample (the per-regime slice of the training stream—the same labelled data the neural decoder sees, so the comparison is fair). Threshold curves sweep p for the seven core decoders under i.i.d. noise, with the spectral decoder calibrated on the i.i.d. stream (where it provably ties the leader); the risk–coverage curve is built from the spectral and matching decoders' pooled confidences. Seeds, platform, library versions, runtime, and peak memory are recorded as provenance in `summary.json`. All numbers in this paper are from the *reported-scale* configuration: distance 9, 20,000 trials per (regime, decoder) (100,000 pooled shots per decoder), 12,000 training/calibration shots, $p_{\text{eval}} = 0.12$, an eleven-point physical-error grid, run end-to-end in 136 s at 262 MB peak memory on a single CPU. A faster smoke configuration (distance 5, 1,500 trials, ≈ 10 s) is provided for continuous integration.

Statistics. Each reported logical error rate is a mean of n independent Bernoulli failure indicators; the ±95% interval is the analytic Wald half-width $1.96\sqrt{\hat{p}(1-\hat{p})/n}$ (Proposition 10). Two decoders are called statistically indistinguishable when their means differ by less than this half-width, and a difference is called significant when the two-sample $z = (\hat{p}_A - \hat{p}_B)/\sqrt{\hat{p}_A(1-\hat{p}_A)/n + \hat{p}_B(1-\hat{p}_B)/n}$ exceeds 3; the per-regime spectral-versus-leader z-scores are written to `summary.json` and audited. The neural classifier's training stream (12,000 shots) is generated independently of the evaluation stream; all randomness derives from a single master seed (`seed` = 0). The exact run provenance—platform `macOS-26.5.1-arm64`, `python` 3.13.12, `numpy` 2.4.3, `scipy` 1.17.1, `networkx` 3.6.1, `scikit-learn` 1.8.0, `matplotlib` 3.10.8—is stored alongside the results. Software: `numpy` [26], `scipy` [30], `networkx` [25], `scikit-learn` [27], `matplotlib` [31]; CPU only.

G. Per-regime results

The per-(regime, decoder) logical error rate, ECE, and coverage at p_{eval} are written in full to `summary.json` under `by_regime`, and the per-regime spectral-versus-leader reductions and z-scores under `headline.spectral_by_regime`. At reported scale the qualitative pattern is regime-dominated for the agnostic decoders: i.i.d. noise yields ≈ 0.002 logical error; biased ≈ 0.043; burst ≈ 0.17; correlated ≈ 0.21; measurement-error ≈ 0.34 (the hardest regime, since syndrome read-out flips corrupt the very signal the decoders condition on). The spectral decoder departs from this cluster on the three structured data-noise regimes (0.024 biased, 0.120 burst, 0.162 correlated) and rejoins it under i.i.d. and read-out noise. Coverage varies strongly by decoder: matching's coverage at the fixed operating threshold ranges from 0.28 (biased) to 0.67 (i.i.d.); topology and the RL selector also abstain substantially; spectral, majority, lookup, neural, and fused commit fully (coverage 1.0) in every regime.

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DATA AND CODE AVAILABILITY

This study uses no external datasets. All numerical values, tables, and figures are generated by the accompanying scripts from fixed random seeds and written to a single source-of-truth artifact (`results/summary.json`); they are reproduced by running the pipeline. The code that implements every code, noise model, decoder, metric, and the audit gate, and that regenerates every figure, table, and reported number, accompanies this manuscript in `submission/code` and is intended for public release upon publication. All experiments are reproducible on commodity hardware; runtime and memory are reported for each benchmark.

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Appendix A: Supplementary information: a from-first-principles construction

This appendix is a self-contained derivation of every concept the accompanying code implements, written to be read top-to-bottom by someone who has seen linear algebra and basic probability but *not* quantum error correction, stabilizer codes, minimum-weight matching, belief propagation, calibration, or selective prediction. Every symbol used in the code is defined, and each subsection names the source file (under `submission/code/src/topoqec/`) it maps to, so theory and implementation can be read side by side. It expands on, rather than repeats, the Methods (Sec. V); where a definition already appears there, we cross-reference it.

1. The problem in one paragraph

A quantum error-correcting (QEC) code stores one logical bit redundantly across many physical bits. Noise flips some physical bits; we may not look at the bits directly (that would destroy the quantum state), so instead we measure a few parity checks whose outcomes form a *syndrome*. A *decoder* takes the syndrome and guesses a correction. It succeeds if (correction $\oplus$ error) is a do-nothing operator (a stabilizer) and fails if that residual flips the logical bit; the usual figure of merit is the *logical error rate*. The central point of this work is that the textbook and learned decoders all optimise the *i.i.d.* likelihood (minimum Hamming weight), so on a repetition code they are statistically *tied*—and the way to do better is to model the *structure* of the noise. We introduce a channel-adaptive *spectral* decoder (a truncated transfer operator on the defect graph) that does exactly that, beating the i.i.d.-optimal leader by 20–44% under biased, bursty, and correlated noise while honestly tying it where no advantage is possible. We further argue that two axes usually ignored both matter: *selective decoding*—letting the decoder *abstain* when unsure, so a slower fallback can take over—and *calibration*—whether the decoder's stated confidence matches how often it is right (the spectral decoder's confidence is the exact model posterior, so it is calibrated by construction). We build one benchmark that measures all three (logical error, calibration, risk–coverage) for nine decoders across five structured-noise regimes, with honest 95% confidence intervals and per-regime significance tests.

2. Codes, syndromes, and logical failure (`codes.py`)

Repetition code (`RepetitionCode`). The primary benchmark. A logical bit is encoded in d data bits over $\mathbb{F}_2 = \{0, 1\}$ (the field of integers mod 2; XOR is "+"). The code has $d-1$ weight-two parity checks $Z_i Z_{i+1}$. An error is a vector $e \in \mathbb{F}_2^d$ ($e_i = 1$ means data bit i is flipped), with syndrome

$$s_i = e_i \oplus e_{i+1}, \qquad i = 0, \ldots, d-2 \qquad (\texttt{.syndrome()}). \tag{A1}$$

Intuitively s_i fires exactly where an error *chain* crosses link i, so the lit syndrome bits sit at the *endpoints* of error chains; decoding is a matching problem on a one-dimensional line of links. The single logical observable is the global parity $L(e) = \left(\sum_i e_i\right) \bmod 2$ (`.logical_value()`). A correction $\hat{c}$ is a *logical failure* (`.logical_error()`) unless the residual $r = e \oplus \hat{c}$ both (a) reproduces zero syndrome (r is a stabilizer, acting trivially) and (b) has zero logical value; a correction that does not reproduce the measured syndrome is automatically a failure. This is the precise definition of "the code protected the bit" used in Methods (Sec. V A).

Lookup table. For small d we enumerate all 2^d errors (`.all_errors()`) and store, for every syndrome, the minimum-weight error that produces it—the coset leader. This optimal table is the gold-standard decoder for small d and the yardstick everything else is compared against (Sec. A 5, `LookupDecoder`).

Surface-like code (`SurfaceLikeCode`). A second, two-dimensional code: $d \times d$ data bits on a grid with $(d-1)^2$ weight-four plaquette checks; a lit plaquette is a *defect*, and defects are paired by Manhattan (L_1) distance. The logical observable is the parity of the left column. It exposes the same defect geometry in 2D so the matching and topology decoders generalise, but it is a topology-*faithful* toy, *not* a stabilizer-exact CSS surface code—which we state honestly (Sec. IV A). Both codes expose a bipartite Tanner graph via `networkx` (`.tanner_graph()`) so graph-based decoders operate on the code topology directly.

3. Structured noise: why i.i.d. is not enough (`noise.py`)

Real hardware noise is *not* independent and identically distributed. Five seeded samplers, each taking a `numpy Generator` and returning a 0/1 error vector, model the structure that matters (`noise.sample()` dispatches on the regime name; see Sec. V B for the parameters):

- `iid`: each bit flips independently with probability p (`sample_iid`).
- `biased`: even-index bits flip at $\min(1, \text{bias} \cdot p)$, odd-index at p (default bias = 3)—a coarse Pauli-bias surrogate (`sample_biased`).
- `burst`: scanning the line, with probability p flip a contiguous run of length $1 + \text{Poisson}(\ell - 1)$, with $\ell =$ burst_len (default 3)—spatially clustered errors (`sample_burst`).
- `correlated`: an i.i.d. seed pattern recruits each neighbour with probability corr (default 0.5) over two sweeps—neighbour correlations (`sample_correlated`).
- `measurement_error`: i.i.d. data noise at rate p *plus* a per-check syndrome read-out flip mask at rate q (default 0.05); the runner XORs that mask into the *measured* syndrome, so the decoder conditions on a corrupted signal (`sample_measurement_error`).

These are precisely the regimes where topology- or read-out-aware decoding might beat a memoryless majority vote—and they turn out to dominate the difficulty far more than the choice of decoder (Sec. A 9).

4. What a decoder returns: the unified interface (`decoders.py`)

Every decoder implements one interface,

$$\texttt{decode(syndrome)} \longrightarrow (\text{correction, confidence}), \quad \text{(A2)}$$

with correction in $\mathbb{F}_2^n$ and confidence in $[0, 1]$. The single interface is what makes the nine decoders comparable on one footing and what makes abstention (Sec. A 7) and calibration (Sec. A 8) definable uniformly. Two helpers recur: `_chain_from_defects()` turns a set of paired defect endpoints into a flip chain, and the defect-fraction confidence heuristic

$$\texttt{_defect_confidence}(n_{\text{def}}, n_{\text{checks}}) = e^{-2\, n_{\text{def}}/n_{\text{checks}}} \quad \text{(A3)}$$

makes a decoder less sure as more checks light up.

5. The decoder family: textbook to learned (`decoders.py`)

- `LookupDecoder`: the optimal minimum-weight coset leader from the enumerated table; confidence boosted toward 1 since the leader is exact. The gold standard for small d.
- `MajorityDecoder`: cumulative XOR (prefix sum mod 2) of the syndrome recovers a flip chain up to a global offset; pick the lower-weight of the two cosets; confidence is the normalised majority margin.
- `MatchingDecoder`: minimum-weight perfect matching of defects. On the line, pair lit syndrome bits left-to-right (with a boundary closure for an odd count) and return the lighter coset. On the surface, build a complete graph on lit plaquettes weighted by negative Manhattan distance and call `networkx max_weight_matching` (Edmonds' blossom algorithm [9]), then flip an L-shaped path between matched pairs. Confidence is the defect-fraction heuristic, damped when a defect is unpaired. This is the standard surface-code decoder, in toy form.
- `BPDecoder`: a few rounds of min-sum belief propagation on the Tanner graph with log-likelihood priors; if the thresholded beliefs do not reproduce the syndrome, defer to matching. (BP underpins quantum LDPC decoding [10, 12].)
- `NeuralDecoder`: a learned two-stage coset selector. Stage 1: a syndrome-determined base chain (always syndrome-consistent). Stage 2: a `scikit-learn LogisticRegression` that predicts the logical coset (0/1) of the true correction relative to that base chain. The output is therefore always syndrome-valid and the confidence is the predicted-class probability—which is exactly why it tends to be well calibrated (Sec. A 8). It is fit on a mixed-regime training stream.
- `TopologyDecoder`: matching, but with confidence *recalibrated* by defect-geometry features (count, span, gap structure).
- `FusedDecoder`: matching and neural both propose; agreement boosts confidence, disagreement defers to the more confident proposal with damped confidence (confidence fusion).
- `RLDecoder`: a contextual bandit (reinforcement learning). Context = a coarse defect-count bucket; actions = candidate decoders. It keeps per-context running success estimates, plays ε-greedy during a warm-up stream, and at evaluation picks the highest-value decoder per context.
- `SpectralDecoder`: the channel-adaptive contribution. It `fit()`s a truncated transfer operator $P_\theta(e) = \hat{\pi}_0(e_0) \prod_i \hat{T}_i(e_i, e_{i+1})$ to a calibration sample of the

operating channel (each $\hat{T}_i$ a normalised, Laplace-smoothed 2×2 count matrix), then on each syndrome compares the model log-likelihood of the two coset representatives $\{c_0, c_0 \oplus \mathbf{1}\}$ and returns the more probable one. Confidence is the exact model posterior $\sigma(|\Delta\ell|)$. Unlike the eight structure-agnostic decoders, it does not target minimum Hamming weight; it targets maximum probability under the *learned* channel, which is why it beats the leader under structured noise (Theorem 6) and reduces to it under i.i.d. noise.

`build_decoders()` wires the family together for a given code; the same list is summarised in Sec. V C.

6. Metric 1: logical error rate and honest intervals (`metrics.py`)

For each (regime, decoder) we Monte-Carlo many shots and record a per-shot failure indicator $F \in \{0,1\}$. The logical error rate is $\hat{p} = \text{mean}(F)$ (`logical_error_rate()`). Because F is Bernoulli, its sample standard deviation is $s = \sqrt{\hat{p}(1-\hat{p})}$ and a 95% confidence-interval half-width is

$$\text{ci}_{95} = 1.96\,\frac{s}{\sqrt{n}} \qquad \texttt{(summarize())}, \tag{A4}$$

with n the number of shots. This is the whole point of honesty: two decoders whose means differ by *less* than a CI width are *not* separated at 95%. At reported scale ($n = 100{,}000$ pooled shots) the eight structure-agnostic decoders span only 0.1508–0.1542 with $\text{ci}_{95} \approx 0.0022$, so their 0.0033 best-to-worst spread is below 1.5 interval widths: they are statistically *tied* (Sec. III B). The channel-adaptive spectral decoder, by contrast, sits at 0.1291—about ten interval widths below the cluster—so its advantage is real, not sampling noise.

7. Metric 2: selective decoding and the risk–coverage curve (`metrics.py`)

A decoder inside a fault-tolerant loop need not commit to every syndrome: it can *abstain* (flag a shot unreliable) so a fallback handles it—the QEC analogue of *selective prediction* in machine learning [20, 22]. Fix a confidence threshold τ and commit only on shots with confidence $\geq \tau$. Then

$$\text{coverage}(\tau) = \text{fraction of shots committed}, \tag{A5}$$

$$\text{risk}(\tau) = \text{logical error rate on committed shots}. \tag{A6}$$

Sweeping τ from the smallest to the largest observed confidence traces the *risk–coverage curve* (`risk_coverage_curve()`), reported in descending-coverage order. As we abstain more (coverage shrinks), risk should fall: we drop the shots we were least sure about. `is_monotone_risk_coverage()` programmatically verifies risk is (weakly) non-increasing as coverage shrinks. Empirically the effect is large: matching's committed error falls from 0.1525 at full coverage to 0.0744 at ≈ 0.54 coverage and 0.0108 at ≈ 0.25 coverage (Sec. III C). The formal reason is Theorem 12: under perfectly calibrated confidence C (so $\Pr(\text{success} \mid C = c) = c$), the committed risk is $1 - \mathbb{E}[C \mid C \geq \tau]$, which can only decrease as τ rises—but *only* to the extent that confidence is calibrated. Hence Sec. A 8.

8. Metric 3: calibration (expected calibration error) (`metrics.py`)

A confidence score is useful for abstention only if it is *honest* about its own accuracy. The expected calibration error (`expected_calibration_error()`) partitions the shots into ten equal-width confidence bins; within each it compares the mean stated confidence to the mean realised accuracy (= 1−failure), takes the absolute gap, and averages those gaps weighted by bin occupancy:

$$\text{ECE} = \sum_{\text{bins}} \frac{n_{\text{bin}}}{n} \left|\overline{\text{conf}}(\text{bin}) - \overline{\text{acc}}(\text{bin})\right|. \tag{A7}$$

ECE = 0 means "when the decoder says 0.8, it is right 80% of the time". Two findings live here. Across the eight structure-agnostic decoders, ECE spans an eight-fold range (0.0446–0.2624) and is *uncorrelated* with sophistication (Sec. III D)—it must be *measured*, not assumed from accuracy or complexity. The spectral decoder is the interesting case: because its confidence is the exact model posterior (Proposition 8), its ECE is near zero wherever the transfer operator matches the channel (0.00069 i.i.d., 0.00159 biased, 0.02538 burst—the best in those regimes by orders of magnitude) and large only under read-out noise (0.32359), where it conditions on a corrupted syndrome. Its pooled ECE (0.0629) is thus competitive with, but not better than, the best agnostic decoders—a number the per-regime view (Table IV) explains exactly.

9. Structured noise sets the difficulty; channel-awareness sets the winner (`runner.by_regime`)

The per-(regime, decoder) breakdown shows the noise regime to be the dominant axis of *difficulty* (Sec. III E, Sec. V G): logical error rises $\approx$ 100-fold from ≈ 0.002 (i.i.d.) to ≈ 0.043 (biased), ≈ 0.17 (burst), ≈ 0.21 (correlated), and ≈ 0.34 (measurement error, the hardest—read-out flips corrupt the very signal decoders condition on). Within each regime the eight agnostic decoders cluster inside their (now tight) CIs. The spectral decoder is the exception: it is the strictly lowest entry on the i.i.d., biased, burst, and correlated columns (0.00145, 0.02400, 0.11960, 0.16190), beating the leader by $z > 10$ on the three structured data-noise regimes, and rejoins

the cluster only under read-out noise. Coverage varies strongly: matching, topology, and RL abstain heavily under structure, whereas spectral, majority, lookup, neural, and fused commit fully.

10. Integrity: automorphism invariance and the audit gate (`symmetry.py`, `audit.py`)

Automorphism invariance. The repetition code is invariant under cyclic relabeling of its data bits (a ring symmetry). `cyclic_relabel(error, shift)` rotates an error; `logical_error_rate_under_relabeling()` and `is_automorphism_invariant()` check that the matching decoder yields the *same* logical error rate on rotated errors (over shifts $\{1, 2, d/2\}$, within a statistical tolerance). A faithful decoder must respect the code's symmetry; the run exposes this as the `automorphism_invariant` integrity flag (true at scale; Sec. V E).

Audit gate. `audit.py` enforces scientific hygiene: `audit_numbers()` checks every claimed number traces to a value in the generated `summary.json` (the single source of truth); `audit_reproducibility_phrase()` checks the mandated reproducibility statement is present; and `audit_forbidden_claims()` rejects unevidenced superlatives. The driver `scripts/audit_claims.py` additionally requires the central spectral-versus-leader advantage to be statistically significant ($z > 3$) in at least one structured regime and not significant under i.i.d. noise, runs in continuous integration, and exits non-zero on any violation—so the paper cannot drift from the artifact.

11. The single source of truth and how to reproduce (`runner.py`)

`runner.run()` ties it together: build the code; fit the neural decoder on a mixed-regime training stream; warm up the RL bandit on its own Monte-Carlo stream; for each (regime, decoder) re-fit the spectral decoder's transfer operator to that regime's calibration sample, then Monte-Carlo `trials` shots at p_{eval} and record per-shot failure and confidence; sweep p over a grid (i.i.d.) for the threshold curves; build the risk–coverage curve from the spectral and matching confidences; check automorphism invariance. It writes `results/summary.json`—the single source of truth from which every table, figure, and reported number is generated. Nothing is typed by hand. Then:

- `scripts/make_tables.py`: `summary.json` → `results/main_results.tex` and `results/paper_tables.tex` (`booktabs`).
- `scripts/make_figures.py`: `summary.json` → the four figure PDFs `fig_schematic`, `fig_perregime`, `fig_threshold`, and `fig_risk_coverage`, as vector PDFs with embedded editable text (`pdf.fonttype=42`), sans-serif mathtext, the colour-blind-safe Okabe–Ito palette, despined axes, and 95% CI bands.
- `scripts/audit_claims.py`: the readiness gate of Sec. A 10.

To reproduce everything:

```
pip install -e .
export KMP_DUPLICATE_LIB_OK=TRUE
bash scripts/reproduce_all.sh full   # d=9
bash scripts/reproduce_all.sh        # CI smoke
```

Reported run: repetition code, distance 9, five regimes, 20,000 trials/regime (100,000 pooled shots/decoder), 12,000 training/calibration shots, $p_{\text{eval}} = 0.12$, seed 0 (Sec. V F). The figure and table artifacts are copied verbatim into `submission/` to build the paper.

12. What the result means (and what it does not)

The eight structure-agnostic decoders are statistically *indistinguishable* on raw logical error for a distance-d repetition code—as expected, since they all approximate the i.i.d.-optimal coset leader (Theorem 5). The contribution is the decoder that escapes that ceiling by modelling the channel: the channel-adaptive spectral decoder is Bayes-optimal for the structured channel (Theorem 6) and cuts the logical error by 20–44% ($z > 10$) under biased, bursty, and correlated noise—an evidenced, statistically significant win—while honestly tying the leader under i.i.d. and read-out noise, where no single-shot data-channel advantage exists. The separable contributions are thus: (i) the channel-adaptive spectral decoder and its provable optimality and calibration; (ii) the framework—a whole decoder family on one defect-graph interface under structured noise with honest CIs and per-regime significance tests; (iii) selective decoding—abstention cuts committed logical error several-fold for a moderate coverage cost; and (iv) calibration as a first-class metric—ECE spans 8× across the agnostic family and is near-zero for the spectral decoder where its model matches. Limitations are stated plainly in Sec. IV A: toy codes only (repetition code; surface-*like* toy, not a stabilizer-exact surface code), simulated structured noise (not hardware syndromes), a required channel calibration sample, a single distance, a first-order model, and a single author and environment. Extending the transfer-operator decoder to a full surface code at scale, to higher-order and online channel models, to hardware syndromes, and to a distance-scaling (sub-threshold) study—while keeping selectivity and calibration first-class—is the natural next step.